

# Session Objectives

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- 1. Describe the specific goals of congestive heart failure (CHF) management.**
- 2. Correlate the pathophysiology of CHF with the drugs used for treatment.**
- 3. Explain how these drugs benefit the CHF patient.**
- 4. Examine the pharmacological mechanism of these drugs.**
- 5. Describe specific drug adverse effects.**

- Inability of ventricle to pump blood into circulation
- Causes: recent MI, uncontrolled hypertension, arrhythmia.
- Drugs: Doxorubicin (intercalating agent, chemotherapeutic).  
Thiazolidinediones “glitazones” (PPAR  $\gamma$  agonists for type II diabetes). Pioglitazone, Rosiglitazone  
NSAIDs (vasoconstriction in kidney by prostaglandin blockade in the afferent arteriole)
- Results in cardiac remodeling

# Cardiac Remodeling

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- **Post MI**
- **Stretch of muscle fibers**
- **Results in fibrosis, collagen, and myocyte hypertrophy**
- **Consequences: Reduced stroke volume and cardiac output, increased ventricle stiffness, decreased contractility**
- **Increase in sympathetics and renin-angiotensin-aldosterone axis (benefit of beta blockers, ACEI/ARBs/ARNI/spironolactone)**

# CHF

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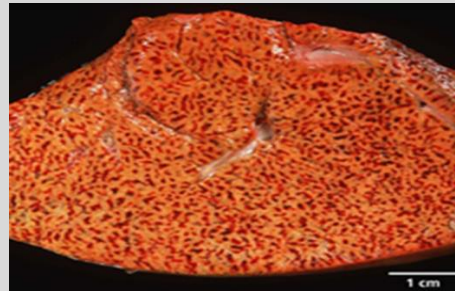
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## Left ventricular failure

- Heart can't maintain body's blood flow needs
- Pressure in pulmonary circulation increases
- Edema, dyspnea, orthopnea, fatigue, hypoxia, organ dysfunction

## Right ventricular failure

- Right ventricle can't pump to lungs. Fluid build-up
- Ankle edema
- Hepatojugular congestion "nutmeg liver"



# CHF Management

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- **Diet, exercise, sodium restriction guidelines**
- **Decrease preload: diuretics, ACEIs/ARBS, venodilators**
- **Decrease afterload: ACEIs/ARBS, arteriodilators**
- **Improve contractility: digoxin, B<sub>1</sub> agonists, PDE3 blockers**
- **Decrease remodeling: ACEIs/ARBS, spironolactone, beta blockers.**
- **Other drugs: Antiplatelet or anticoagulants, antidiabetics, antihyperlipidemics, etc**

# Positive Inotropic Drugs

- Increase contractility, mainly for acute decompensated heart failure
- Digoxin (digitalis, foxglove plant)
- Dobutamine ( $B_1$  agonist,  $G_s$  mechanism).
- Dopamine (maintains GFR and renal blood flow)
- Milrinone (PDE-3 blocker)

# Digoxin

- N/V/D, anorexia
- Ventricular arrhythmias: High Ca, low K, Mg (monitor diuretic use)
- Yellow vision (halos)
- Gynecomastia (estrogenic)
- DI: antacids, cholestyramine (absorption) amiodarone (CYP inhibition), verapamil and quinidine (elimination)
- Renal impairment (reduced digoxin elimination)
- Antidote: DigFab, manage electrolytes. Give magnesium (torsades)



# Vasodilators

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- Angiotensin-converting enzyme inhibitors (ACEI, -prils) and Angiotensin II receptor blockers (ARBs, -sartans). **\*\*Chronic CHF\*\*\*\*\***
- Hydralazine
- Nitric oxide donors (nitroglycerin, isosorbide dinitrate, nitroprusside)

# RAAS Adverse Effects

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- First-Dose Phenomenon
- Aldosterone absence: Hyperkalemia
- **Bradykinin: Cough, angioedema. Specific to ACEI. Replace with ARBs**
- Drug interactions (K<sup>+</sup> sparing diuretics or supplements), NSAIDs.
- Renal artery stenosis (contraindication). But used to **reduce proteinuria in type II diabetics**
- Fetal renotoxicity (not in pregnancy)
- Best ACEI for CHF: ramipril, enalapril, and lisinopril
- Aliskiren: direct renin inhibitor (similar adverse effects)

# Hydralazine



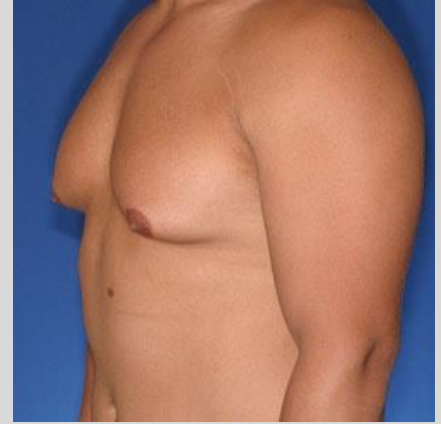
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- Arteriole (decrease afterload)
- Severe hypertension in pregnancy
- Acetylated (procainamide, isoniazid, sulfonamides)
- Adverse effects: headache, flushing, dizziness, hypotension, **tachycardia, fluid retention.**
- **Lupus-like syndrome** (slow acetylators)

# Aldosterone Blockers (MRA)

- Spironolactone, Eplerenone, Finerenone
- Antagonizes aldosterone receptors
- **Spironolactone (androgen blocker)**
- Use: high aldosterone/ CHF, acne, hirsutism
- Adverse effects: Hyperkalemia, acidosis
- Spironolactone: gynecomastia, impotence.
- Drug interactions: Digoxin, “renin” drugs, NSAIDs
- Finerenone: highly selective MRA. Slows progressive of CKD.



# Loop Diuretics

- **Loss of Na, Cl, K, Ca, and Mg**
- **Increases PGE<sub>2</sub> (vasodilation)**
- **Use: CHF, pulmonary edema, hypercalcemia, hyperkalemia, hypertension, acute renal failure**
- **Adverse effects: hypokalemia, hypomagnesemia, hypocalcemia, hyperuricemia, metabolic alkalosis. Ototoxicity (especially with ethacrynic) hypersensitivity (except ethacrynic acid)**
- **Drug interactions: NSAIDs, digoxin, overlap ototoxicity (cisplatin, gentamicin, vancomycin)**

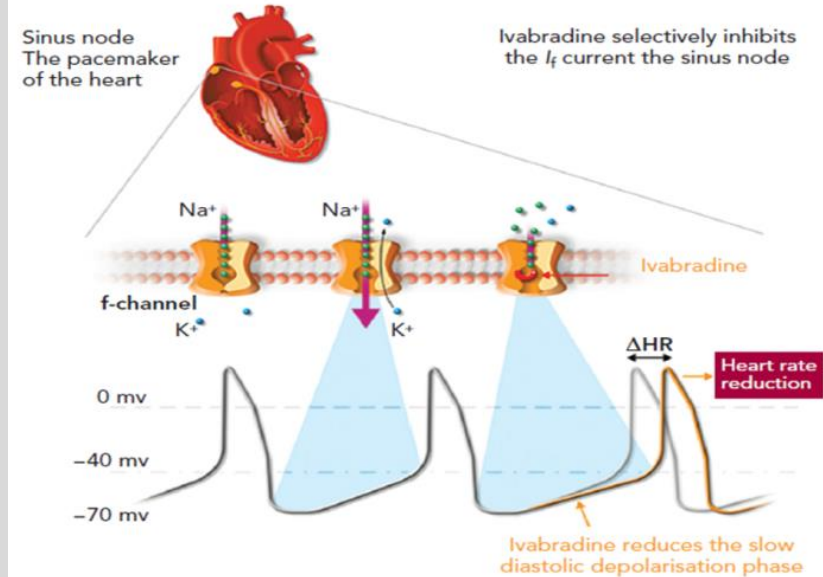
# Thiazide Diuretics

- **Hydrochlorothiazide, Chlorthalidone, Indapamide**
- **Use: hypertension, CHF, kidney stones, and nephrogenic diabetes insipidus**
- **Adverse effects: Hypokalemia, alkalosis, hyperglycemia, hypercalcemia, hyperlipidemia, hyperuricemia, hypovolemia, hyponatremia, sulfa sensitivity.**
- **Drug Interactions: digoxin, NSAIDs, lithium**

# Ivabradine

- Selective  $I_f$  inhibitor (“funny channel” blocker)
- Inhibits action in the SA node without affecting the duration of the action potential
- Use: CHF and stable angina, especially those who can't tolerate or be managed with beta antagonist
- Adverse effects: bradycardia, vision changes, not in pregnancy. Caution with CYP 3A4 inhibitors

Figure 1: Mechanism of Action of Ivabradine



Source: <http://www.shift-study.com/ivabradine/mode-of-action/> Reproduced with the permission of Servier © 2016.

# Other Drugs

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- Angiotensin receptor/neprilysin inhibitor (ARNI): Sacubitril: (prodrug) inhibits neprilysin to enhance natriuretic peptides. Combined with Valsartan. Adverse effect: hypotension, hyperkalemia.
- Sodium-glucose cotransporter 2 inhibitor (SGLT2): dapagliflozin
- Conivaptan: blockade of  $V_{1a}$  receptor decreases PVR.  $V_2$  blockade causes water secretion.
- Semaglutide: obesity related heart failure.

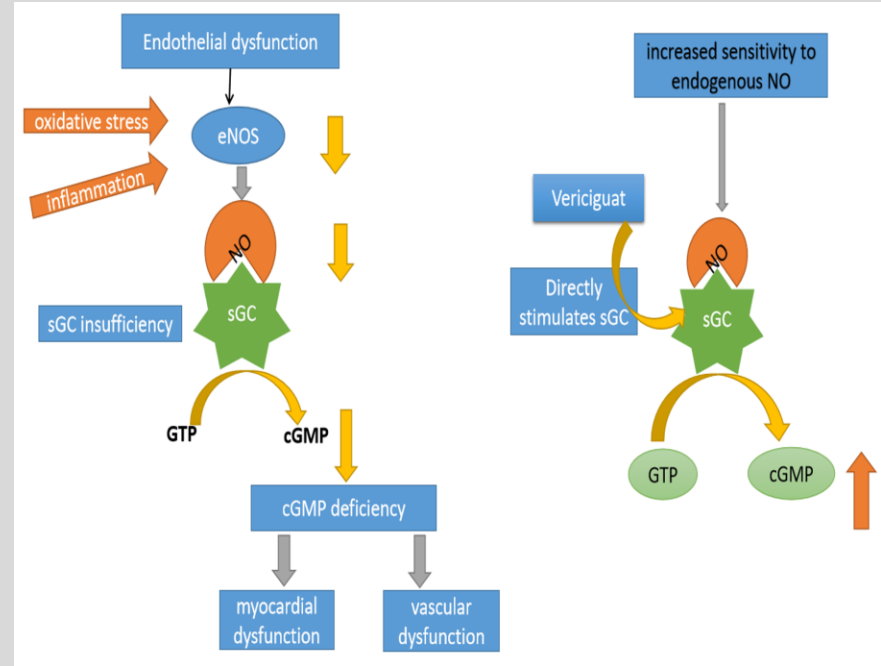
# Subtype 2 sodium-glucose transport protein (SGLT2) inhibitors

- Canagliflozin, **dapagliflozin, empagliflozin**
- Glucose excreted by kidney via proximal tubule
- CHF management: promotes diuresis.
- ~ 1% A1c reduction
- Adverse Effects: UTIs, monitor GFR (not for CrCl <30)

# Vericiguat

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- Stimulates soluble guanylate cyclase that activates the cGMP pathway independent of nitric oxide
- Adverse effects: Headache, dizziness, hypotension, tachycardia. Not to be taken with NO donors or PDE5-
- Does not induce tolerance.
- Reduced hospitalizations in patients with worsened chronic CHF (HFrEF)



# Pharmacologic therapy of HFrEF

- Improvement in symptoms: diuretic, beta blocker, ACEI (or ARB), ARNI, SGLT2-, hydralazine plus nitrate, digoxin, and mineralocorticoid receptor antagonist (MRA).
- Prolongation of patient survival: beta blocker, ACE inhibitors (or ARB), ARNI, dapagliflozin, hydralazine plus nitrate, and MRA.
- Limited evidence of survival benefit is available for diuretic therapy.

# Review Q

**A 66-year-old female was diagnosed with ventricular arrhythmia and started on medication. Several months later, she presented with excessive fatigue, weight gain, hair loss, and dry skin. Laboratory studies revealed a low T4 and a elevated TSH. Which drug induced this presentation?**

- A. Disopyramide**
- B. Lidocaine**
- C. Propranolol**
- D. Amiodarone**
- E. Diltiazem**